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Sophie Eberdold (UT2J)
Marcos Didonet del Fabro (CEA List)
UT2J-IRIT-SM@RT

Subject: Application for PhD Position – Methodology for Engineering Digital Twins (PC3)

Dear Sophie Eberdold and Marcos Didonet del Fabro,

I am writing to express my profound interest in the PhD position on "Methodology for Engineering Digital Twins," co-supervised at UT2J-IRIT-SM@RT and CEA List. With my robust engineering background in systems architecture, Model-Driven Engineering (MDE), and AI integration, I am highly motivated to tackle the challenge of defining rigorous methodologies for the lifecycle management and federation of digital twins.

A central challenge in current digital twin engineering is the lack of standardized formalisms to manage their evolution, orchestration, and validation. During my engineering of the "Autonomous Cyber-Shield Platform" and the "Industrial Gas Turbine Digital Twin," I confronted the complexities of integrating heterogeneous models—from purely physical thermodynamic simulations to data-driven AI inference components. To ensure interoperability and maintainability, I employed strict systems engineering principles, creating modular boundaries that allowed each sub-system to evolve independently while still communicating through standardized interfaces. This hands-on experience in architecting complex, multi-layered cyber-physical systems perfectly positions me to develop the methodological frameworks you require. I am eager to research novel model-driven techniques to define clear workflows, ensure traceability, and automate the deployment of digital twins across the Artemis platform.

Furthermore, my deep technical knowledge enables me to bridge the gap between high-level conceptual modeling and low-level programmatic implementation. I am eager to collaborate closely with the AT3 engineering teams to validate these methodologies on real-world industrial use cases. Please find attached my resume and academic transcripts. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui